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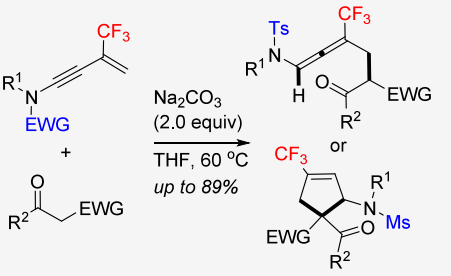
γ-CF_3-Allenamides versus 3-CF_3-Cyclopentenylamines: Substituent-Controlled Divergent Reaction of β-CF_3-1,3-Enynamides with β-Dicarbonyl Compounds

Jintong Li, Yu Wu, Mingqing Liu, Lei Wang, and Yuanjing Xiao*

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ABSTRACT: A distinctive *N*-substituent-controlled regioselective 1,4-hydrocarbonation or 1,4-hydrocarbonation/5-*endo*-*trig* cyclization cascade reaction of β-CF_3-1,3-enynamides with β-dicarbonyl compounds in the presence of a simple base is reported. β-CF_3-1,3-enynamides having a Ts group *N*-substituent produce γ-CF_3-allenamides via a 1,4-hydrocarbonation process, whereas β-CF_3-1,3-enynamides bearing a Ms group *N*-substituent lead to 3-CF_3-cyclopentenylamines through a 1,4-hydrocarbonation/5-*endo*-*trig* cyclization cascade process.



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Allenamides[1] and cyclopentenylamines[2] are two types of important structural motifs of natural products and pharmaceutically relevant molecules (Figure 1). The allenamide motifs are also recognized as flexible synthetic building blocks in diverse synthetic conversions and are outstanding precursors for the synthesis of natural products.[3] Consequently, the synthesis of these two versatile units has garnered extensive research interest, leading to a range of strategies for their functionalized construction.[4,5] On the other hand, building blocks incorporating CF_3 are regarded as essential structural motifs in the fields of medicinal chemistry and agrochemistry.[6] Due to the distinctive reactivity of allenes[7] and their capacity to participate in various transformations to create valuable complex structures,[8] trifluoromethyl-allenes have attracted significant attention regarding their synthetic method. As a consequence, substantial efforts have been made over the years by synthetic organic chemists to design and develop methods allowing efficient access to a variety of functionalized CF_3-allenes.[9] Despite the impressive achievements in CF_3-allenes synthesis, however, less attention has been paid to the development of new synthetic methods for the synthesis of trifluoromethylated allenamides. In 2016, Evano and Blanchard first reported the synthesis of γ-trifluoromethylated allena-

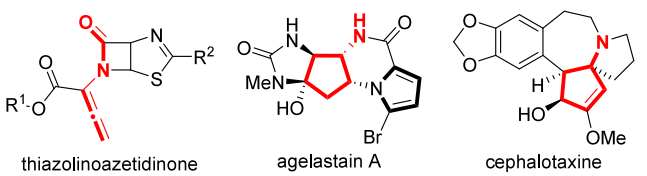
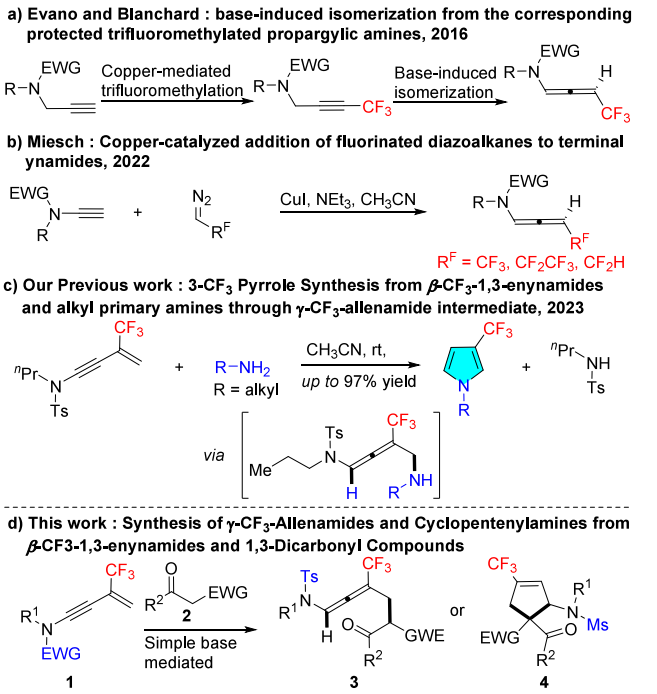


Figure 1. Selected bioactive molecules featuring an allenamide or cyclopentenylamine skeleton.

Scheme 1. Synthetic Approaches to γ-CF_3-Allenamides



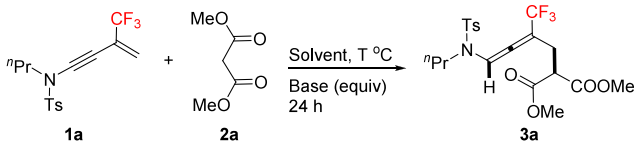
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Table 1. Initial Reaction Discovery and Optimization of Reaction Conditions[a]

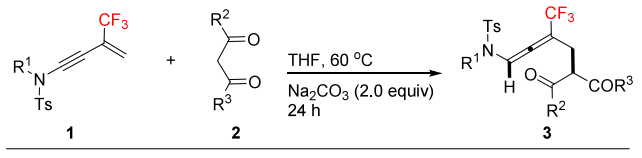


1a 2a 3a

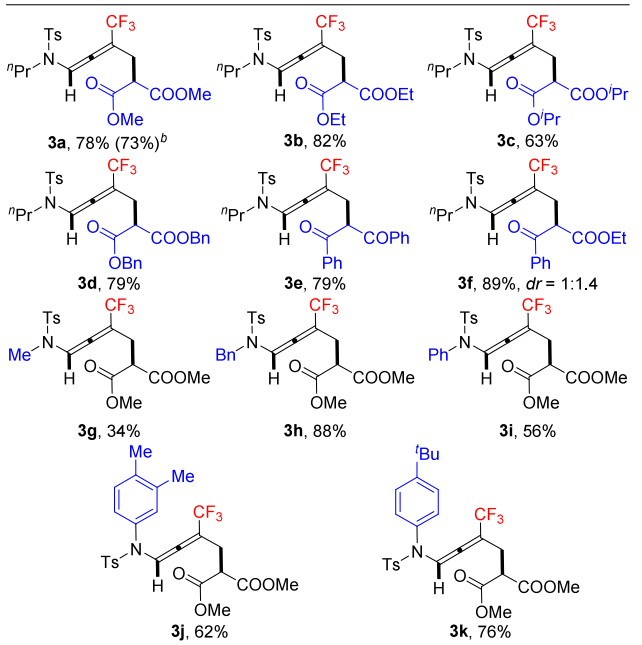
entry	solvent	base (equiv)	1a/2a	T (°C)	3a (%)[b]
1	CH_3CN	Na_2CO_3(1.5)	1:1.5	r.t.	
2	CH_3CN	Na_2CO_3(1.5)	1:1.5	60	33
3	CH_3CN	Na_2CO_3(1.5)	1:1.5	80	n.d.
4	CH_3CN	Cs_2CO_3(1.5)	1:1.5	60	32
5	CH_3CN	Et_3N(1.5)	1:1.5	60	
6	CH_3CN	DMAP(1.5)	1:1.5	60	
7	DMSO	Na_2CO_3(1.5)	1:1.5	60	15
8	THF	Na_2CO_3(1.5)	1:1.5	60	67
9	THF	Na_2CO_3(2.0)	1:3	60	73
10[c]	**THF**	**Na_2CO_3(2.0)**	**1:3**	**60**	**81 (78)**

[a]Reactions were conducted with **1** (0.1 mmol) in solvent (2.0 mL). [b]NMR yields using CH_2Br_2 as the internal reference; yields of isolated product are shown in parentheses. [c]**1a**, 0.5 mmol scale.

Scheme 2. Scope of the Synthesis of CF_3-Allenamides[a]

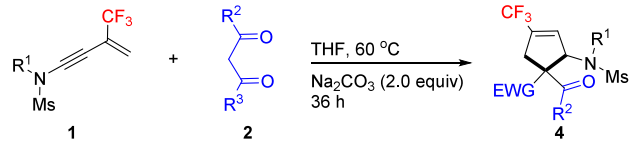


1 2 3

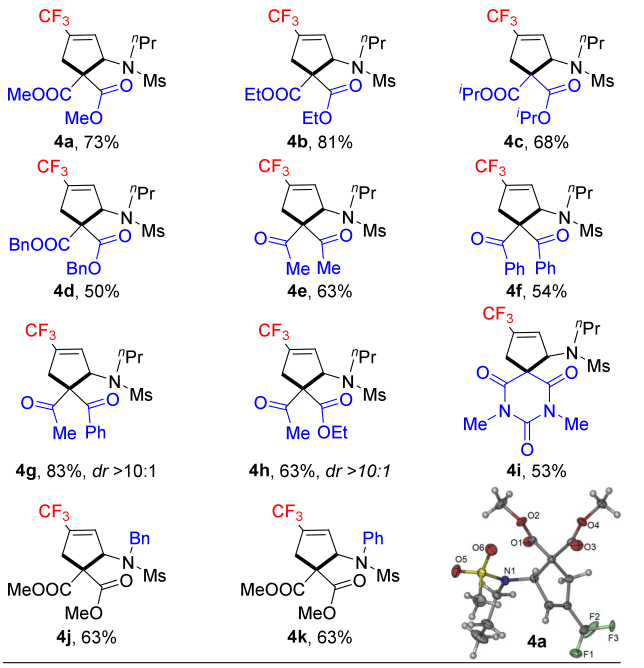


[a]Reactions were conducted with **1** (0.5 mmol) and **2** (1.5 mmol) in THF (2.0 mL). Diastereoselectivity was determined by ^{1}H NMR analysis. [b]1 mmol scale reaction was performed.

Scheme 3. Scope of the Synthesis of Cyclopentenylamines[a]

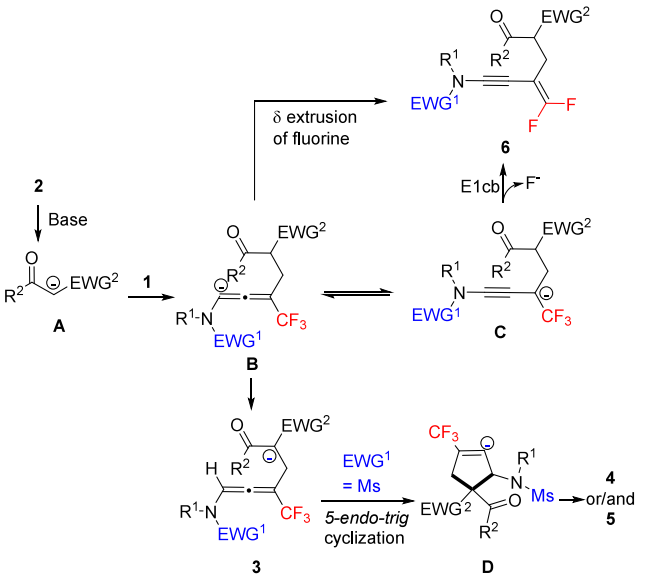


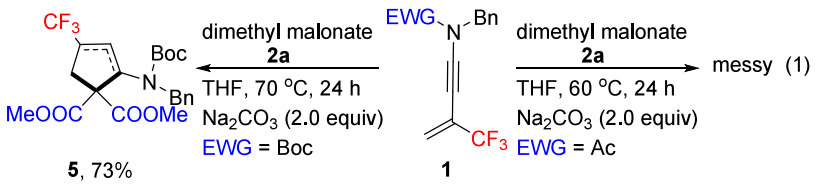
1 2 4



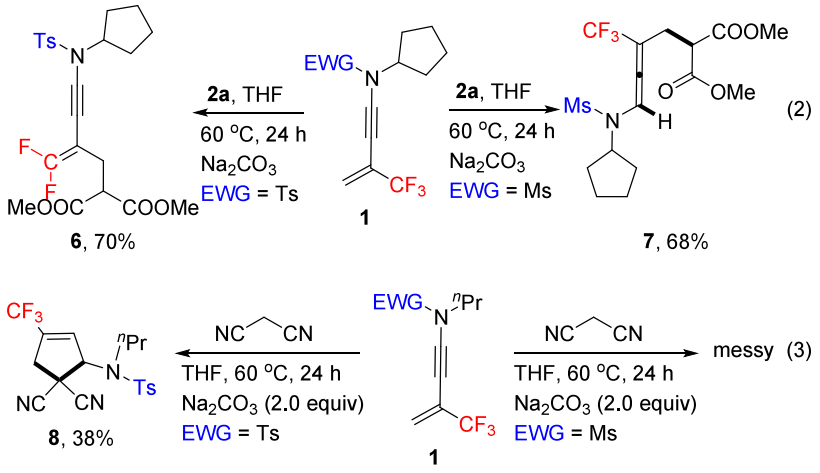
[a]Reactions were conducted with **1h** (0.5 mmol) and **2** (1.5 mmol) in THF (2.0 mL). Diastereoselectivity was determined by ^{1}H NMR analysis.

Scheme 4. Proposed Mechanism

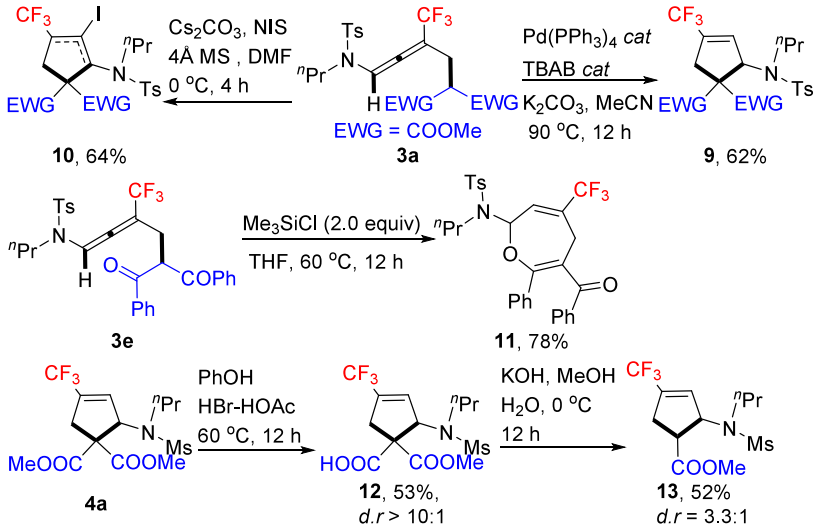




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J. Org. Chem. 2024, 89, 13789−13794



Scheme 5. Synthetic Application



mides from protected trifluoromethylated propargylic amines via base-induced isomerization (Scheme 1a). The exploration of the reactivity of γ-trifluoromethylated allenamides demonstrated that they serve as outstanding substrates for cyclizations catalyzed by radicals and gold(I), leading to the formation of fluoroalkylated nitrogen heterocycles.[10] In 2022, Miesch's group reported an elegant copper-catalyzed coupling of ynamides and fluorinated diazoalkanes, enabling fluorine-substituted *N*-allenamides to be obtained (Scheme 1b).[11] Among them, CF_3-*N*-allenamides are useful building blocks for the preparation of functionalized stereodefined difluorinated dienes, monofluorinated dienes,[12] and γ-sultams.[13] Very recently, we developed a new metal-free method for the efficient preparation of 3-trifluoromethyl pyrroles through highly regioselective 1,4-hydroamination/cyclization/aromatization cascade reactions of β-CF_3-1,3-enynamides with alkyl primary amines. Evidence for the highly regioselective 1,4-hydroamination was observed after the isolation and characterization of the allenamide intermediate (Scheme 1c).[14] As a continuation of our interest in the synthesis of fluorinated carbocycles and heterocycles[15] and the chemistry of β-CF_3-1,3-enynamides, we were intrigued by the possibility of extending this annulation method to include 1,3-dicarbonyl compounds. It is well-known that the electronic properties and the reactivity of allenamides can be tuned by the selection of the *N*-substituent group.[3b] We envisioned that functionalized γ-CF_3-allenamide intermediate **3** would be formed in the same way through regioselective 1,4-hydrocarbonation of β-CF_3-1,3-enynamide **1** with β-dicarbonyl compound **2** under a proper choice of a base. Unlike our previous reports,[14] this γ-CF_3-allenamide intermediate **3** may be stable and separable or undergo further intramolecular 5-*endo*-*trig* cyclization to give 3-CF_3-cyclopentenylamines **4**, depending on the *N*-substituent group. Herein, we are pleased to report our recent results on this simple base-mediated divergent reaction of β-CF_3-1,3-enynamides with β-dicarbonyl compounds leading to γ-CF_3-allenamides or 3-CF_3-cyclopentenylamines, and the reactions are substrate-dependent (Scheme 1d).

To test our assumption, 4-methyl-*N*-propyl-*N*-(3-(trifluoromethyl)but-3-en-1-yn-1-yl)benzenesulfonamide **1a** and dimethyl malonate **2a** were selected as model substrates for reaction screening. Initially, **1a** and **2a** (1.5 equiv) were subjected to the reaction in CH_3CN at room temperature with Na_2CO_3(1.5 equiv) as a base; however, after 24 h, no reaction occurred according to TLC analysis (Table 1, entry 1). To our delight, when the reaction was performed at 60 °C, the expected functionalized γ-CF_3-allenamides **3a** was indeed formed, albeit in 33% yield without full consumption of material **1a** (Table 1, entry 2). To speed up the reaction, an elevated temperature (80 °C) was used; however, the reaction

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J. Org. Chem. 2024, 89, 13789−13794

became complicated, indicating the allenamide product was not stable at high temperature (Table 1, entry 3). To further improve the yield of **3a**, we turned our attention to screening the base and solvent, and Cs_2CO_3 gave a similar result as Na_2CO_3 in terms of yield and conversion (Table 1, entry 4); however, organic bases such as Et_3N and DMAP gave no reaction (Table 1, entries 5 and 6). Among the solvents tested, THF was found to be optimal (Table 1, entries 7 and 8). Moreover, by increasing the ratio of malonate **2a** to β-CF_3-1,3-enynamide **1a**, the yield was further improved (Table 1, entry 9). After numerous attempts, we were pleased to find that the desired γ-CF_3-allenamide **3a** product was isolated in 78% yield when the reaction was run at an increased concentration (Table 1, entry 10), and the reaction conditions were thus defined as the standard reaction conditions for subsequent investigations.

With suitable reaction conditions established (Table 1, entry 10), we next explored the scope and limitations of the process with respect to the β-CF_3-1,3-enynamides bearing a Ts *N*-substituent group and β-dicarbonyl compounds, and the results are summarized in Scheme 2. First, the reaction can be scaled up to 1 mmol without a loss of efficiency. In general, a series of malonate derivatives, β-diketones, and β-ketoesters could be employed as reaction partners to afford the corresponding functionalized γ-CF_3-allenamides in moderate to good yields (**3a**−**3f**). In the case of malonate, the steric effect of the substitution on the ester group could not be negligible; for example, changing the ester group from methyl to isopropyl or benzyl showed an obvious difference in terms of yield (**3a**−**3d**). When the unsymmetrical β-dicarbonyl compounds such as β-keto esters were subjected to the reaction, the desired allenamide **3f** was obtained as a mixture of two diastereomers in good yields but with poor diastereoselectivity (1:1.4). With respect to the β-CF_3-1,3-enynamides with a Ts *N*-substituent group, substituents linked to the nitrogen atom also have dramatic effects on the outcome of the reaction. For example, switching *n*-propyl to a methyl group, the reaction became complicated, some unidentified compounds were formed, and allenamide **3g** was only obtained in 34% isolated yield. However, benzyl was tolerated for this transformation, producing allenamide **3h** in 88% isolated yield. β-CF_3-1,3-Enynamides **1** with Ts *N*-substituent groups bearing different aryl groups such as phenyl and substituted phenyl linked to the nitrogen atom are quite general substrates for the synthesis of γ-CF_3-allenamides (**3i**−**3k**).

In the course of studying the reactivity of β-CF_3-1,3-enynamides **1** through the selection of different electron-withdrawing groups attached to the nitrogen atom, we found it very interesting that under similar reaction conditions the replacement of the Ts group in β-CF_3-1,3-enynamides **1** with a Ms group switches the reaction outcome, and cyclopentenylamine formed via double hydrocarbonation sequence was observed (Scheme 3). For example, treatment of *N*-propyl-*N*-(3-(trifluoromethyl)but-3-en-1-yn-1-yl)methanesulfonamide **1h** with dimethyl malonate **2a** under similar reaction conditions for γ-CF_3-allenamide synthesis delivers cyclopentenylamine **4a** in 73% isolated yield. The structure of **4a** was further confirmed by single-crystal X-ray analysis.[16] We think that allenamide intermediates with Ms groups formed via the first intermolecular hydrocarbonation have less steric effects and thus are prone to the second intramolecular hydrocarbonation to give the cyclopentenylamine product. The scope of β-dicarbonyl compounds was then examined, and the results are summarized in Scheme 3. Similar to that for γ-CF_3-allenamide formation, a series of malonate derivatives, β-diketones, β-ketoesters, and *N*,*N*-dimethylbarbituric acid are quite general substrates for cyclopentenylamine synthesis (**4a**−**4i**). Similarly, when the unsymmetrical β-dicarbonyl compounds such as unsymmetrical 1,3-diketones or β-ketoesters were subjected to the reaction, the desired cyclopentenylamines were obtained as a mixture of two diastereomers in good yields with good diastereoselectivity (**4g**, **4h**). β-CF_3-1,3-Enynamides **1** with a Ms *N*-substituent group bearing other alkyl group such as benzyl or an aryl such as phenyl linked to the nitrogen atom are also quite general substrates for cyclopentenylamine synthesis (**4j**, **4k**).

The scope of the β-CF_3-1,3-enynamides **1** with other electron-withdrawing groups such as Ac and Boc was also investigated using dimethyl malonate **2a** as the reaction partner. β-CF_3-1,3-Enynamides with an Ac *N*-substituent group failed to give any detected products, and the reaction became very complicated; however, the reaction of β-CF_3-1,3-enynamides with a Boc *N*-substituent group needs an elevated temperature and affords inseparable double bond isomer mixture of cyclopentenylamine **5** in 73% isolated yield (eq 1). Interestingly, β-CF_3-1,3-enynamides with a Ts *N*-substituent group bearing cyclopentyl linked to the nitrogen atom affords gem-difluorinated ene−ynamides **6** through an S_N2' reaction process[17] in 70% isolated yield; however, replacement of the Ts group with Ms group switches the reaction outcome, and functionalized γ-CF_3-allenamide **7** is formed (eq 2). Besides β-dicarbonyl compounds, active methylene compounds such as malonitrile were also investigated. β-CF_3-1,3-Enynamides with a Ts or Ms *N*-substituent group bearing *n*-propyl linked to the nitrogen atom both fail to undergo a clean reaction with malonitrile. β-CF_3-1,3-Enynamides with the Ms *N*-substituent group failed to give any detected products; through replacement of the Ms group with the Ts group, cyclopentenylamine **8** could be obtained, albeit in low yield (eq 3). These results clearly demonstrated that both substituents on the nitrogen atom *of* β-CF_3-1,3-enynamides and nucleophiles had a dramatic effect on the reaction diversity.A plausible mechanism for the divergent formation of allenamides, cyclopentenylamine, and gem-difluorinated ene−ynamides is proposed in Scheme 4. The regioselective nucleophilic addition of the anion intermediate **A** to β-CF_3-1,3-enynamide **1** would afford two resonance-stabilized anions, allenamide carbanion intermediate **B** or propargylic anion intermediate **C** (α-trifluoromethyl carbanion), depending on the *N*-substituent group effects of enynamide **1** and nucleophiles. The stabilized allenamide carbanion would facilitate allenamide **3** formation. If allenamide **3** contained suitable electronic and steric factors, it would further undergo an intramolecular nucleophilic addition under basic reaction conditions to give the cyclopentene carbanion intermediate **D**. The protonation of intermediate **D** would produce the final product cyclopentenylamine **4**. On the other hand, if an α-trifluoromethyl carbanion is prone to form, it would undergo defluorination to give gem-difluorinated ene−ynamide **6** via an E_1cb reaction process. An alternative reaction pathway for gem-difluorinated ene−ynamide **6** formation from allenamide carbanion intermediate **B** through δ-extrusion of fluorine was also plausible.[12]

To demonstrate the synthetic practicability of this methodology, further transformations representative of γ-CF_3-allenamides **3a** and **3e** and cyclopentenylamine **4a** were performed (Scheme 5). A Pd-catalyzed[18] or NIS-mediated[19]

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cycloisomerization of γ-CF_3-allenamide **3a** furnished the cyclopentenylamine **9** and an inseparable double bond isomer mixture of iodo-substituted cyclopentenylamine **10**, respectively. Upon treatment of γ-CF_3-allenamide **3e** with 2.0 equiv of TMSCl in THF at 0 °C, the trifluoromethyl-substituted 2,5-dihydrooxepine **11** could be obtained in 78% isolated yield. In addition, one ester group at the ring position of cyclopentenylamine **4a** could be hydrolyzed under acidic reaction conditions by treatment of **4a** with HBr-HOAc at 60 °C, affording cyclopentenylamine **12** with the Ms group intact. Decarboxylation of **12** could be realized under basic reaction conditions to give cyclopentenylamine **13** in 52% yield.

In conclusion, a simple base-mediated 1,4-hydrocarbonation or 1,4-hydrocarbonation/5-*endo-trig* cyclization cascade reaction of β-CF_3-1,3-enynamides with β-dicarbonyl compounds affording γ-CF_3-allenamides or cyclopentenylamines has been developed. The reaction is substrate-dependent. In general, β-CF_3-1,3-enynamides having a Ts group *N*-substituent produce γ-CF_3-allenamides via a 1,4-hydrocarbonation process, whereas β-CF_3-1,3-enynamides bearing a Ms group *N*-substituent lead to 3-CF_3-cyclopentenylamines through a 1,4-hydrocarbonation/5-*endo-trig* cyclization cascade process. Both substituents on the nitrogen atom of β-CF_3-1,3-enynamides and nucleophiles had a dramatic effect on the reaction diversity.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c01669.

Experimental procedures; characterization data; ^1H, ^{13}C and ^{19}F NMR spectral data; mass spectrometry data; and crystal data (PDF)

Accession Codes

CCDC 2365484 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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